

Md Rahatul Ashakin

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Queens, New York, USA · ashakin.rahatul@gmail.com · [Website](#) · [Google Scholar](#) · [ORCID](#) · [GitHub](#) · [LinkedIn](#)

Research Focus

Reliable and efficient machine learning for health, finance, and secure data systems, with an emphasis on

- Optimization for robustness/calibration and deployment,
- Quantum-inspired methods (QAOA/VQE, QML), and
- Privacy/trust on cloud-GPU and edge/IoT systems.

Research and Professional Experience

Data Science Intern, Tekurai Inc., San Antonio, Texas (remote, Queens - NY)

May 2024–Present

- Built transformer and CNN pipelines for clinical text and audio, analyzed clinician and patient interactions for decision support.
- Processed 500k+ healthcare records with Python and produced interactive Power BI dashboards for operations and care insights.
- Prototyped privacy aware blockchain workflows for secure data exchange in drug discovery across multiple institutions.
- Quantum machine learning with Qiskit, including hybrid variational models and quantum guided hyperparameter search that improved internal accuracy and AUC over classical baselines.

Program Manager, ShopUp, Dhaka, Bangladesh

Feb 2022–Sep 2022

- Led analytics for B2C and B2B operations with forecasting and KPI reporting; delivered Tableau and Power BI dashboards used to reduce returns and improve throughput.
- Partnered with operations, product, and engineering to define a KPI taxonomy and the supporting SQL/ETL pipelines; automated weekly executive dashboards and cohort analyses to speed up decision cycles.

Project Manager, Icon Information Systems Ltd., Dhaka, Bangladesh

Jun 2020–Feb 2022

- Directed cross-team reporting and SQL data workflows for enterprise clients; coordinated agile ceremonies and risk tracking.
- Produced project plans and stakeholder updates, organized sprints and stand-ups, and prepared UAT and release checklists to ensure on-time, auditable deliveries.

Assistant Project Manager, Millennium Information Solution Ltd., Dhaka, Bangladesh

Nov 2018–May 2020

- Supported delivery of HRIS and financial systems with user acceptance testing, reliability checks, and transaction safety reviews for regulated stakeholders.

Education

M.S., Information Technology, Washington University of Science and Technology (WUST), Alexandria, Virginia

Oct 2022–Mar 2024

B.S., Computer Science and Engineering, North South University, Dhaka, Bangladesh

Sep 2013–Aug 2018

Publications & Scholarly Contributions

Book Chapter (forthcoming)

- **Ashakin, M. R.** (2026). Chapter 5: *Current Tools and Techniques of Artificial Intelligence in Healthcare*. In D. Roosan (Ed.), *Quantum Computing in Medicine: Transforming Drug Discovery, Protein Folding, and Genomics Research through Artificial Intelligence Integration*. Elsevier. ISBN: 978-0-443-34193-9. *CRedit*: Conceptualization; Methodology; Writing: original draft; Visualization; Writing: review & editing.

Under Review (Journal)

- **Ashakin, M. R.**, Stuetzle, C., Khan, R., Nirzhor, S., Roosan, D. *Quantum Annealing of Financial Sentiment Analysis by Optimizing Large Language Model Hyperparameters*. Under review at *Financial Innovation (SpringerOpen)*. *CRedit*: Lead: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing: original draft. Supporting: Data curation; Writing: review & editing. *Resources*: Code: [GitHub repository](#) (Manuscript available upon request).

Refereed Conference Papers

- Roosan, D., **Ashakin, M. R.**, Khan, R., Hasiba Mashed Khan, Khou, T., Carnasciali, M. I., Haider, M. R. (2025). *Variational Quantum Circuits for Molecular Classification Using Graph Neural Network*. In: *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2025)*. CRediT: Methodology; Software; Formal analysis; Visualization; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T., Nirzhor, S., Haider, M. R. (2025). *Quantum Variational Transformer Model for Enhanced Cancer Classification*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering* CRediT: Methodology; Software; Validation; Formal analysis; Writing: review & editing.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T. (2025). *Adaptive Multimodal Artificial Intelligence with Liquid Neural Network for Edge Computing-Based Augmented Reality*. In: *IEAI 2025, IOS Press, Advances in Transdisciplinary Engineering*. CRediT: Conceptualization; Methodology; Software; Visualization; Writing: review & editing.
- Roosan, D., Khou, T., Pham, B., Li, Y., **Ashakin, M. R.**, Hasiba Mashed Khan, Khan, R., Haider, M. R. (to appear, Nov 2025). In: *Quantum AI Enhanced Blockchain Security for Drug Discovery*. Proceedings of ICAI 2025, Springer LNEE (to appear). CRediT: Conceptualization; Methodology; Writing: review & editing.

Other Peer-Reviewed Articles

Additional peer-reviewed articles (2023–2024): 8 papers in AI for healthcare, IoT, and DSP. Full list: [Google Scholar](#).

Skills

Methods & Algorithms	Transformers · CNNs · GNNs · multimodal learning · optimization for robustness · model calibration · privacy-aware modeling
Data & Experimentation	Dataset curation · experimental design · cross-validation · statistical testing · ablation studies · error analysis · reproducible pipelines
Software & Systems	Python · SQL · Git · Linux · Jupyter · Pandas · NumPy · scikit-learn · PyTorch · Hugging Face
Visualization & Reporting	Matplotlib · Tableau · Power BI · results dashboards · academic writing with LaTeX/Overleaf
Computing Platforms	GPUs · cloud · edge/IoT deployment

Selected Research Projects (open-source)

- **IBD-Quantum-ESM**. Quantum-informed *IBD* modeling pipeline (optimization with classical + quantum methods). *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).
- **Quantum Optimization Sandbox**. Lightweight scaffold to *repurpose* ML experiments with QAOA/VQE and classical baselines. *Resources*: Code: [GitHub](#) · [README](#) · [Notebooks](#).

Online Courses & Certifications (selected)

- [Microsoft Power BI Data Analyst Specialization](#) · **Microsoft**
- [Google Advanced Data Analytics](#) · **Google**
- [Google Business Intelligence](#) · **Google**
- [Google Data Analytics](#) · **Google**
- [SAS Statistical Business Analyst](#) · **SAS**
- [Project Management & Other Tools for Career Development](#) · **UC Irvine**

Academic Service

Research Programmer, Merrimack College (Remote, Queens–NY)

Sep 2024–Dec 2024

Supported *DSE 4900 (Capstone)* and *DSE 3010 (ML for DS)*; coordinated labs and weekly clinics; supervised theses end-to-end (scoping, curation, modeling, reports). Instituted reproducible practice (clean repos, env pinning, experiment tracking) and rigorous evaluation (ablations, error analysis, leakage/imbalance checks), enabling teams to reach *up to ~50% relative accuracy gains* over course baselines where applicable.